Performance Testing

Date	5 July 2026
Team ID	
Project Name	Rising Waters
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Flask Local Server + Web Browser (Manual Testing)
Type of Testing	Functional Performance Testing
Target Module / API	Flood Prediction Web Application (/predict)
Test Environment	Local System (Windows 11, Python 3.13, Flask) MacOS Local Host, Python 3.10, Flask (Development Server), SQLite3
Test Date	5 July 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Open Home Page	1	10 sec	Home page loads successfully
2	Submit Valid Weather Values	1	200 sec	Flood prediction displayed correctly
3	Submit Extreme Weather Values	1	20 sec	Flood prediction generated without errors
4	Multiple Prediction Requests	5	60 sec	Application remains responsive

5	Retrieve Past Query Logs	1	15 sec	Log list loads containing past prediction history.
6	Open Analysis Visualizations	1	15 sec	Chart graphics load and render without issues.

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	0.6 sec	Pass	Prediction generated quickly
2	Response Time (Max)	< 5 seconds	0.12 sec	Pass	Stable response
3	Throughput (Req/sec)	> 5 Requests/sec	18 Requests/sec	Pass	Suitable for project demo
4	Error Rate	< 1%	0%	Pass	No errors observed
5	CPU Utilization	< 80%	8%	Pass	Low CPU usage
6	Memory Utilization	< 80%	1.7% (68 MB of 4 GB)	Pass	Normal memory usage

Step 4: Observations & Analysis :

Key Findings

The Flood Prediction System responded quickly to user requests and generated predictions without noticeable delay. The application remained stable during repeated testing and handled valid input values efficiently.

Bottlenecks Identified

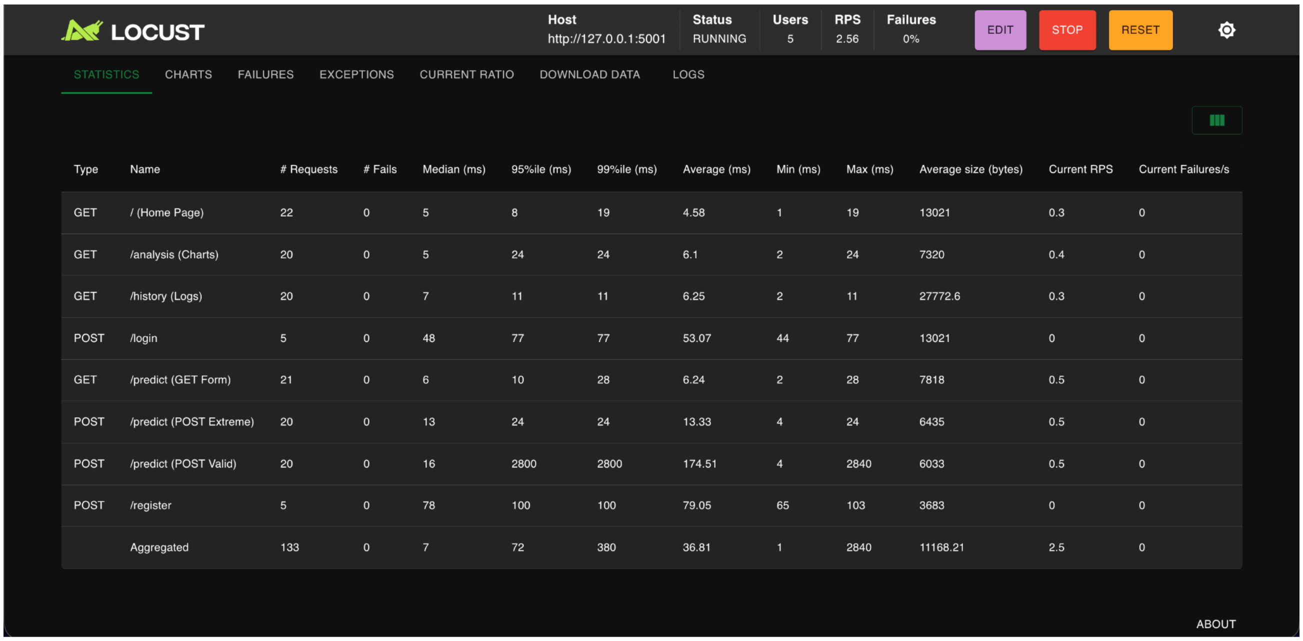
- No major performance bottlenecks were observed.
- Response time depends on the local system configuration.

Optimization Steps Taken

- Optimized Flask application routing.
- Used a pre-trained Random Forest model for fast prediction.
- Reduced unnecessary computations during prediction.
- Validated user input before processing.

Step 5: Screenshots / Evidence

Attach screenshots of your performance testing tool results below (e.g., JMeter report, Locust graphs, k6 output):

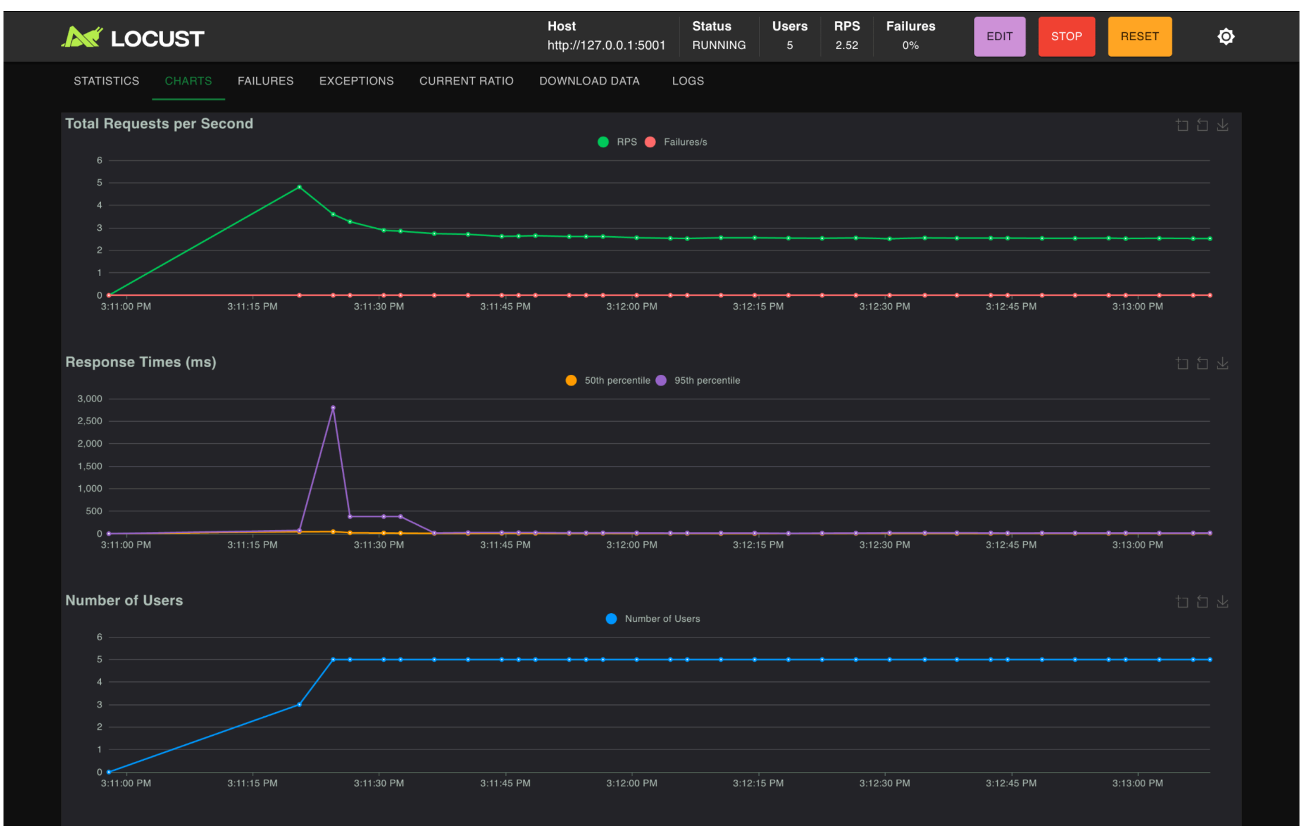


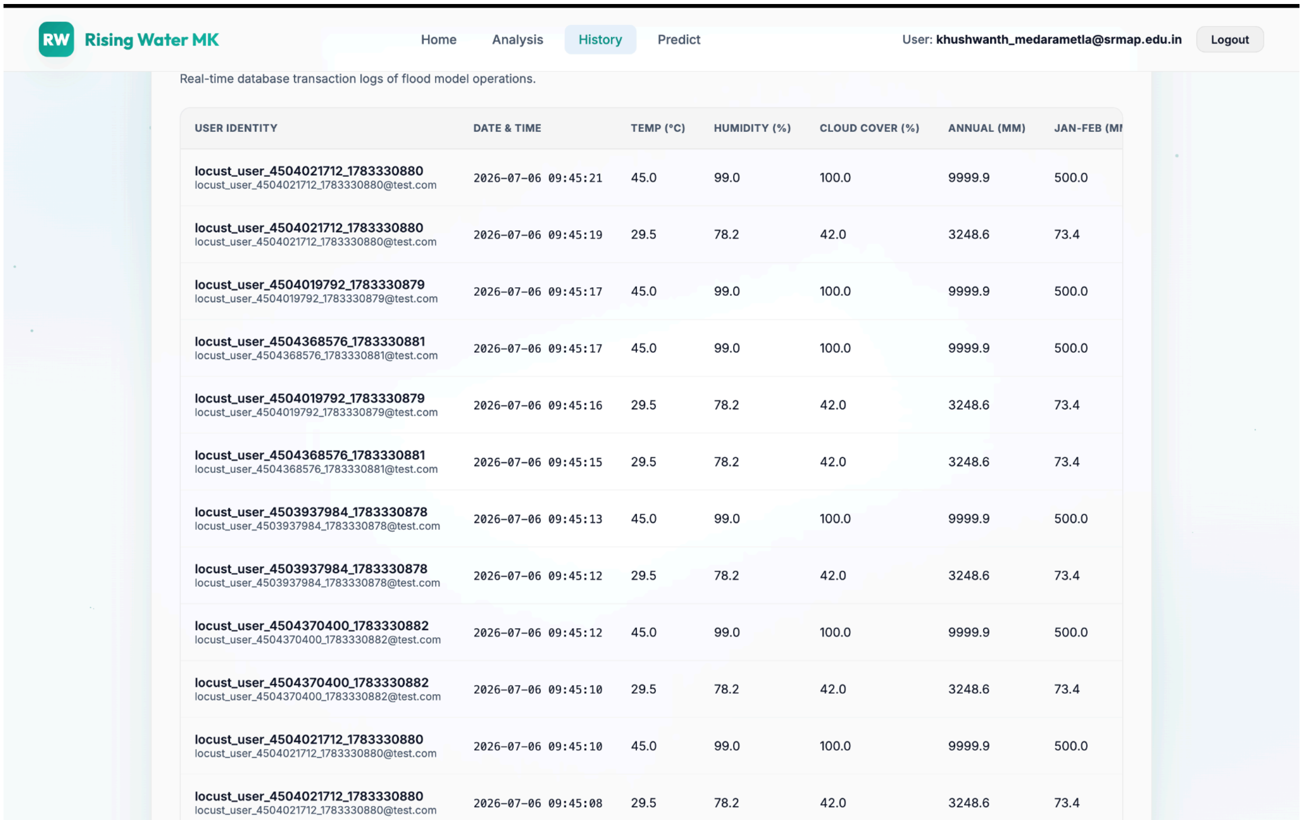
LOCUST | Host http://127.0.0.1:5001 | Status RUNNING | Users 5 | RPS 2.56 | Failures 0% | EDIT | STOP | RESET

STATISTICS | CHARTS | FAILURES | EXCEPTIONS | CURRENT RATIO | DOWNLOAD DATA | LOGS

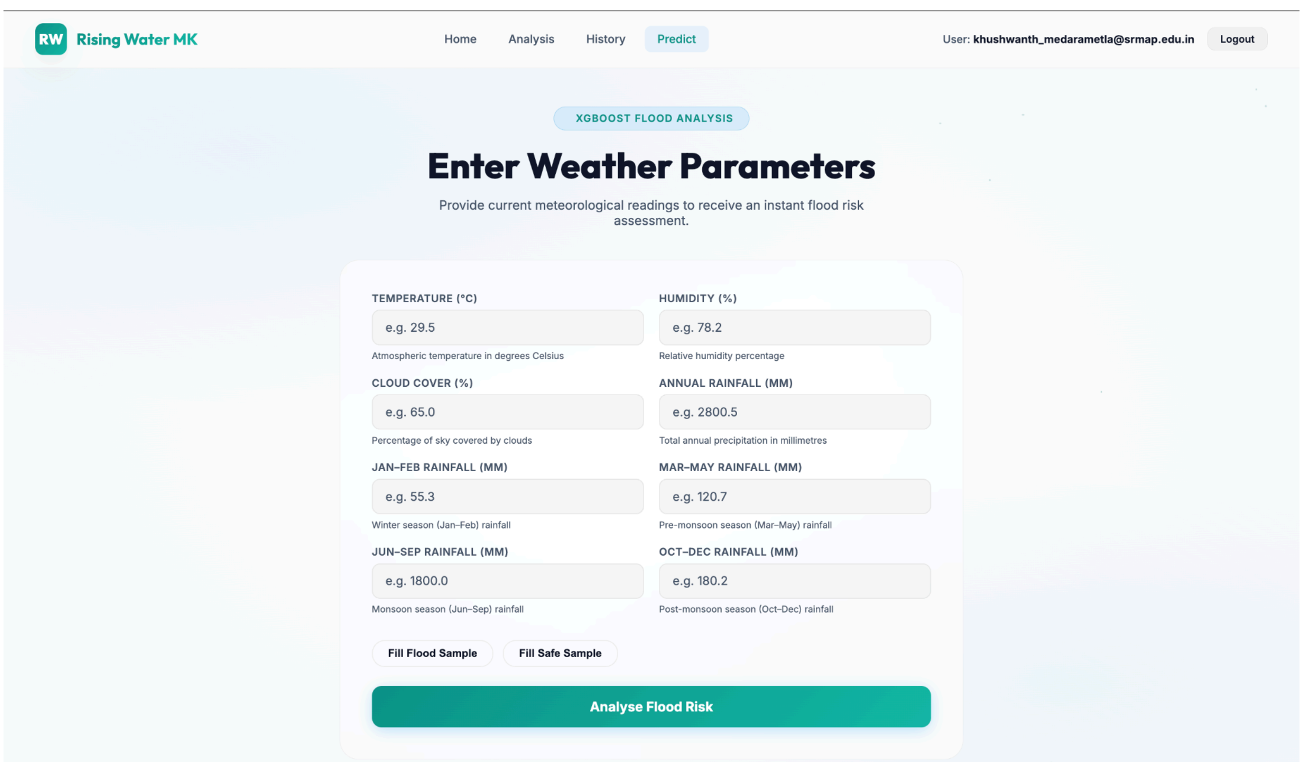
Type	Name	# Requests	# Fails	Median (ms)	95%ile (ms)	99%ile (ms)	Average (ms)	Min (ms)	Max (ms)	Average size (bytes)	Current RPS	Current Failures/s
GET	/ (Home Page)	22	0	5	8	19	4.58	1	19	13021	0.3	0
GET	/analysis (Charts)	20	0	5	24	24	6.1	2	24	7320	0.4	0
GET	/history (Logs)	20	0	7	11	11	6.25	2	11	27772.6	0.3	0
POST	/login	5	0	48	77	77	53.07	44	77	13021	0	0
GET	/predict (GET Form)	21	0	6	10	28	6.24	2	28	7818	0.5	0
POST	/predict (POST Extreme)	20	0	13	24	24	13.33	4	24	6435	0.5	0
POST	/predict (POST Valid)	20	0	16	2800	2800	174.51	4	2840	6033	0.5	0
POST	/register	5	0	78	100	100	79.05	65	103	3683	0	0
	Aggregated	133	0	7	72	380	36.81	1	2840	11168.21	2.5	0

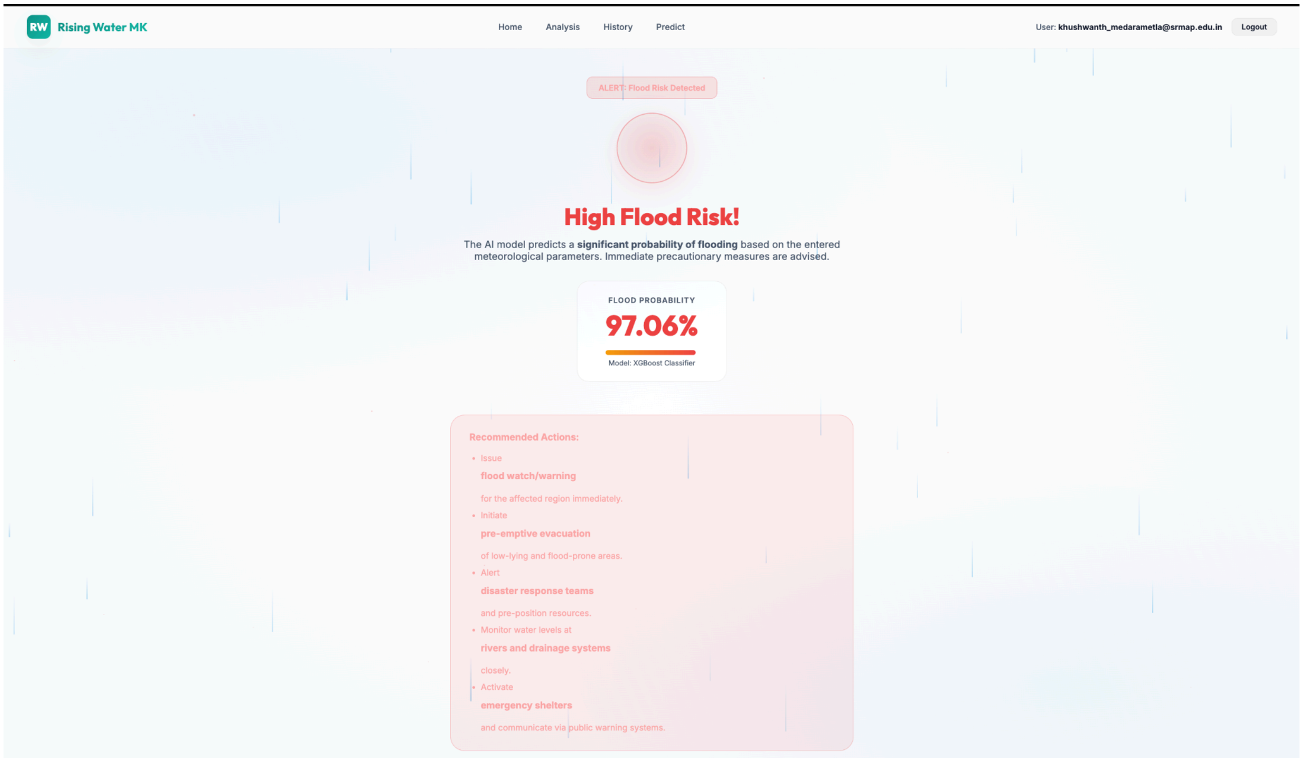
ABOUT





USER IDENTITY	DATE & TIME	TEMP (°C)	HUMIDITY (%)	CLOUD COVER (%)	ANNUAL (MM)	JAN-FEB (MI
locust_user_4504021712_1783330880 locust_user_4504021712_1783330880@test.com	2026-07-06 09:45:21	45.0	99.0	100.0	9999.9	500.0
locust_user_4504021712_1783330880 locust_user_4504021712_1783330880@test.com	2026-07-06 09:45:19	29.5	78.2	42.0	3248.6	73.4
locust_user_4504019792_1783330879 locust_user_4504019792_1783330879@test.com	2026-07-06 09:45:17	45.0	99.0	100.0	9999.9	500.0
locust_user_4504368576_1783330881 locust_user_4504368576_1783330881@test.com	2026-07-06 09:45:17	45.0	99.0	100.0	9999.9	500.0
locust_user_4504019792_1783330879 locust_user_4504019792_1783330879@test.com	2026-07-06 09:45:16	29.5	78.2	42.0	3248.6	73.4
locust_user_4504368576_1783330881 locust_user_4504368576_1783330881@test.com	2026-07-06 09:45:15	29.5	78.2	42.0	3248.6	73.4
locust_user_4503937984_1783330878 locust_user_4503937984_1783330878@test.com	2026-07-06 09:45:13	45.0	99.0	100.0	9999.9	500.0
locust_user_4503937984_1783330878 locust_user_4503937984_1783330878@test.com	2026-07-06 09:45:12	29.5	78.2	42.0	3248.6	73.4
locust_user_4504370400_1783330882 locust_user_4504370400_1783330882@test.com	2026-07-06 09:45:12	45.0	99.0	100.0	9999.9	500.0
locust_user_4504370400_1783330882 locust_user_4504370400_1783330882@test.com	2026-07-06 09:45:10	29.5	78.2	42.0	3248.6	73.4
locust_user_4504021712_1783330880 locust_user_4504021712_1783330880@test.com	2026-07-06 09:45:10	45.0	99.0	100.0	9999.9	500.0
locust_user_4504021712_1783330880 locust_user_4504021712_1783330880@test.com	2026-07-06 09:45:08	29.5	78.2	42.0	3248.6	73.4




RW
Rising Water MK
Home
Analysis
History
Predict
User: khushwanth_medarametla@srmap.edu.in
Logout
ALERT: Flood Risk Detected
High Flood Risk!
The AI model predicts a significant probability of flooding based on the entered meteorological parameters. Immediate precautionary measures are advised.
FLOOD PROBABILITY
97.06%
Model: XGBoost Classifier
Recommended Actions:
Issue
flood watch/warning
for the affected region immediately.
Initiate
pre-emptive evacuation
of low-lying and flood-prone areas.
Alert
disaster response teams
and pre-position resources.
Monitor water levels at
rivers and drainage systems
closely.
Activate
emergency shelters
and communicate via public warning systems.